EEG-Based Cognitive State Classification for Real-Time Car Simulation Control Using Deep Learning

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***Abstract*—Abstract—This research explores the use of electroencephalogram (EEG) signals for identifying cognitive states and interacting with a real-time driving simulator. A deep learning brain–computer interface (BCI) model is developed to classify mental states Focused, Unfocused, and Drowsy using raw EEG data. Using one-dimensional convolutional neural networks (Conv1D), the model was trained on an openly available dataset that was collected using the Emotiv EPOC headset. The proposed model achieves an accuracy of 82.31 and macro F1 score of 0.79 on unseen data. The predicted mental states were used to control the actions of the simulated car body, resulting in dynamic, adaptive control based on cognitive information. This research advances prior EEG-based models that used brain data classification by demonstrating an end-to-end, real-time large-scale solution for continuously measuring cognitive state as a form of interaction in an interactive context. Index Terms—EEG; Brain–Computer Interface; Deep Learning; Cognitive State Classification; Car Simulation; Human–Machine Interaction**

***Index Terms*—EEG, Brain–Computer Interface, Deep Learning, Cognitive State Classification, Car Simulation, Human–Machine Interaction**

I. Introduction

The capability to track and react to human cognitive behavior in real time has a variety of applications in fields such as autonomous driving, neurorehabilitation, gaming, and human–computer interaction. Electroencephalography (EEG), which captures the electrical activity of the brain, provides a non-invasive, fast-acting means of assessing mental states broadly characterized as attention, relaxation, and fatigue. Brain–computer interface (BCI) systems utilize such signals to create direct communication channels between the human brain and external devices, bypassing traditional input methods.

In driving, cognitive fatigue and inattention are two major factors that can contribute to traffic accidents and lower the safety of operation. Current driver-monitoring systems rely heavily on behavioral indicators, such as eye movement, steering deviations, and facial expressions, that may not correlate with underlying cognitive states. EEG-based approaches, on the other hand, provide a more objective and responsive measure of neural activity in real time.

In this paper, we propose a deep learning framework to classify three cognitive states of a driver—Focused, Unfocused, and Drowsy—using multichannel EEG data. The classification is made in real time and the output of the classifier is coupled with a simulated vehicle so that the movements of the car are noticeable with respect to the detected mental state. A publicly available EEG data set will be utilized to analyze different cognitive states in the simulation.

The main contributions of this research are as follows:

- A Conv1D-based neural network architecture is developed for end-to-end classification of EEG signals.
- The trained classifier is included in a real-time vehicle simulation that responds to cognitive predictions.
- Performance comparison with state-of-the-art models demonstrates the potential of the proposed framework for the development of neuroadaptive systems.

II. Literature Review

In recent years, EEG signal-based cognitive state classification research has garnered considerable attention. Lawhern et al. introduced EEGNet, a compact convolutional neural network model designed for EEG-based brain-computer interfaces (BCIs), with effective performance on motor imagery and visual evoked potential tasks [1]. Roy et al. [2] investigated the use of EEG signals for attention monitoring in driving simulators, and affirmed the system's potential for real-time assessment of driver state. Nijholt reviewed BCI application in gaming, discussing how cognitive input can contribute to user engagement [3].

Bird et al. [4] explored EEG-based classification of mental states with the Muse headset, defining calm, neutral, and intense states, and derived 2100+ features including statistical moments and entropy metrics, achieving 87.16

Wearable EEG systems, such as Emotiv and Muse, has also made EEG signal elicitation and real-time disregulation more convenient for users. Sharma et al. [7] compared Emotiv and Neuphony headsets in terms of EEG signal stability for BCI applications, suggesting that the Emotiv headset produce more stable EEG signal activity than the Neuphony headset. Furthermore, Bitbrain [8] has also demonstrated an EEG-based fatigue detection system in industry. Previous work, however, overwhelmingly emphasizes feature engineering and neglects to incorporate real-time control mechanisms. Our work addresses this gap by utilizing band-filtered raw EEG in a Conv1D architecture for real-time simulation control..

III. Problem Statement and Motivation

Driver distraction and cognitive fatigue are significant contributors to traffic collisions. Most past monitoring systems were focused on behavioral indicators and overlooked neural markers of cognitive states. This research proposes to analyze and classify driver cognitive states in real-time using direct interpretations of EEG signals, and will use these predictions to control a behavioural simulation of a driver in a simulated vehicle. The main goals are as follows:

The objectives are:

- Develop a high-performance deep learning model to classify driver states of Focused, Unfocused, and Drowsy.
- Integrate EEG-based cognitive state detections with a simulation to alter real-time behaviour of a simulated vehicle.
- Assess model performance and compare results to the previous body of work.

IV. Dataset Description

The dataset for this research is accessible on Kaggle and was collected with the Emotiv Epoc headset, a 14-channel EEG device at a sampling frequency of 128 Hz [11]. The data consists of EEG samples and are labeled as Focused, Unfocused, and Drowsy, with each sample comprising a 4-second epoch of EEG samples for all channels.

A. Data Composition

- Focused: 5100 samples
- Unfocused: 14218 samples
- Drowsy: 5100 samples

With the class imbalance, training could be problematic due to a model bias towards the majority class. Therefore, stratified sampling was used for train-validation-test splits, and class weighting was used during training.

B. Signal Format

Each EEG sample is stored as a multichannel time series of 512 time steps (4 seconds, 128 Hz) with 14 spatial channels. The raw time series signals are incorporating microvolt units and include EEG characteristics (e.g., baseline drift, muscle artifacts, spikes due to blinking).

V. Preprocessing

EEG signals are noisy and non-stationary by nature. Preprocessing is very important for improving fidelity and finding meaningful patterns.

A. Bandpass Filtering

We used a fourth order Butterworth bandpass filter to separate four canonical frequency bands:

- Delta (0.5 – 4 Hz)
- Theta (4 – 8 Hz)
- Alpha (8 – 13 Hz)
- Beta (13 – 30 Hz)

The filter transfer function is given by:

$$H(s) = \frac{1}{\sqrt{1 + \left(\frac{s}{\omega_c}\right)^{2n}}} \tag{1}$$

where ω_c is the cutoff frequency, and n is the order of the filter.

B. Epoch Segmentation

Each EEG sample was segmented into 4 seconds:

$$N = f_s \cdot T = 128 \cdot 4 = 512 \text{ samples} \tag{2}$$

This segmentation preserves short-term temporal dynamics while preserving computational feasibility.

C. Artifact Removal

To reduce noise from eye blinks and movements, the component analysis was performed using independent components analysis (ICA), to remove components correlated with the frontal channels (AF3, AF4). Additionally, notch filtering at 50 Hz was applied to reduce the power line interference.

VI. Methods of Feature Extraction

Our model operates on raw band-filtered EEG input, but we also built a baseline using handcrafted features in a different experimental condition.

A. Statistical Features

For all channels and bands, we calculated the mean, standard deviation, skewness, and kurtosis of the signal. In general, these features pick up distributional characteristics of the signal.

B. Entropy Features

We calculated the Shannon entropy and log-energy entropy to assess the complexity and randomness of the signal:

$$H = -\sum_{i=1}^{N} p_i \log(p_i) \tag{3}$$

C. Frequency Domain Features

We computed the Fast Fourier Transform (FFT) of each epoch which identifies the spectral power of the bands. Measures of peak frequency and band ratios (e.g., alpha/beta) were used as cognitive features.

D. Temporal Derivatives

We took first order and second order derivatives of the signal to assess transient shifts and power dynamics of slope.

VII. Deep Learning Structures for EEG

There are several deep learning structures have been proposed for the classification of EEG:

A. EEGNet

EEGNet [1] is a compact CNN structure for EEG-based BCIs that takes advantage of depthwise and separable convolution to reduce the number of parameters and to maintain spatial-temporal structure.

B. CNN-LSTM Hybrids

Nandihal et al. [**?**] proposed hybrid structures that took advantage of CNNs for spatial feature extraction and LSTMs for temporal modeling, which achieved high accuracy in classifying cognitive states.

C. Transformer Models

Recent work has investigated transformer-based models for EEG, capitalizing on self-attention mechanisms to capture long-range dependencies. However, these models need large datasets and can be computationally expensive.

VIII. The Proposed Model: Conv1D Architecture

We constructed a sequential neural network based on Conv1D to learn spatiotemporal features from EEG data.

The architecture consists of:

- Input Layer: Shape = (512, 14)
- Conv1D Layer 1: 32 filters, kernel size = 5, ReLU activation
- MaxPooling Layer 1: Pool size = 2
- Dropout Layer 1: Rate = 0.3
- Conv1D Layer 2: 64 filters, kernel size = 3
- MaxPooling Layer 2: Pool size = 2
- Dropout Layer 2: Rate = 0.3
- Flatten Layer
- Dense Layer: 64 units, ReLU activation
- Output Layer: 3 units, Softmax activation

A. Softmax Output

The model produced a probability distribution over classes using:

$$P(y_i) = \frac{e^{z_i}}{\sum_{j=1}^{3} e^{z_j}} \tag{4}$$

B. Loss Function

The model was optimized with categorical cross-entropy loss:

$$L = -\sum_{i=1}^{3} y_i \log(\hat{y}_i) \tag{5}$$

We trained with the Adam optimizer using early stopping and a model checkpoint to save based on validation accuracy.

IX. Training Configuration and Hyperparameter Tuning

The Conv1D model proposed in this study was constructed using Python and TensorFlow. Training took place on a workstation with an NVIDIA RTX 3060 GPU (12GB VRAM) and 32GB RAM. The dataset was divided into training (70%), validation (15 percent), and testing (15 percent) sets utilizing stratified sampling to maintain class distribution.

A. Hyperparameters

The following hyperparameters were determined through static and empirical tuning based on a grid search:

- Number of epochs = 50
- Batch size = 64
- Learning rate = 0.001 (with ReduceLROnPlateau)
- Optimizer = Adam
- Loss function = Categorical Crossentropy
- Early stopping = Patience = 5 (monitored on validation loss)

B. Regularization

To reduce overfitting, dropout layers were applied after every convolutional and dense layer. Regularization using L2 was also tried but seemed to have minimal impact relative to dropping out neurons. Batch normalization was also considered, however, it was ruled out due to increased training time with no increase in performance.

X. Evaluation Metrics

To assess model performance, we used the following metrics:

A. Accuracy

The overall classification accuracy is defined as:

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}} \tag{6}$$

B. Precision, Recall, and F1-Score

These metrics provide class-wise insights:

$$\text{Precision} = \frac{TP}{TP+FP}, \quad \text{Recall} = \frac{TP}{TP+FN}, \quad F1 = \frac{2 \cdot \text{Precision} \cdot \text{R}}{\text{Precision} + \text{Re}} \tag{7}$$

C. Confusion Matrix

The confusion matrix visualizes prediction errors across classes.

D. Macro F1-Score

Macro-averaging treats all classes equally:

$$\text{Macro F1} = \frac{1}{C}\sum_{i=1}^{C} F1_i \tag{8}$$

where C represents the number of classes.

XI. Experimental Results

The model was evaluated on a test set of 4884 samples. Key results include:

- Test Accuracy: 84.93%
- Macro F1-score: 0.81
- Training Time: 4 minutes per epoch

TABLE I
CLASS-WISE PRECISION, RECALL, AND F1-SCORE

Class	**Precision**	**Recall**	**F1-Score**
Focused	0.82	0.89	0.85
Unfocused	0.91	0.91	0.91
Drowsy	0.70	0.64	0.67
Accuracy			84.93%
Macro Avg	0.81	0.81	0.81
Weighted Avg	0.85	0.85	0.85

TABLE II
CONFUSION MATRIX

True / Pred	**Focused**	**Unfocused**	**Drowsy**
Focused	906	26	88
Unfocused	64	2591	189
Drowsy	141	228	651

XII. INTEGRATION WITH CAR SIMULATION

A vehicle simulation in Pygame was developed to analyse an EEG-predicted car control in real time.

- **Focused:** Speed = 120 km/h
- **Unfocused:** Speed = 40 km/h
- **Drowsy:** Speed = 0 km/h (Crash occurred at 10 predictions in a row)

The system integrates EEG signal processing, deep learning-based cognitive state classifications, and real-time car control via a simulation engine. The system consists of three core modules:

1) **EEG Acquisition and Preprocessing:** EEG signals were acquired via the Emotiv Epoc headset. The raw EEG signals went through preprocessing steps including bandpass filters and segmentation to separate meaningful epochs of EEG data from which to analyse.
2) **Deep Learning Model:** A 1D Convolutional Neural Network (Conv1D) was used to classify states of cognition such as focus, relaxation, or mental workload. The deep learning model was trained on preprocessed EEG data and optimized for low-latency inference.
3) **Car Simulation Engine:** A Pygame-based simulation environment receives cognitive state predictions and translates these predictions into controls (e.g., acceleration, braking, and steering). This provides real-time interaction and feedback.

Intermodule communication is executed using shared memory buffers and a socket interface for ease of low-latency data handling and modularity. This architecture facilitates seamless integration between brain-signal interpretation and autonomous control systems.

A Pygame-based car simulation was built to visualize real-time control using EEG predictions. The simulation dynamically adjusts the vehicle's speed according to the predicted cognitive state, providing immediate visual feedback on the driver's alertness level.

This approach not only demonstrates the feasibility of translating cognitive state predictions into actionable control signals but also provides an intuitive platform for testing and refining brain-computer interface algorithms in a safe, controlled environment.

The system's modular design allows for future enhancements, such as incorporating additional control parameters (e.g., steering angle, braking force), improving prediction accuracy through multimodal sensor fusion, and extending the framework to real vehicle control scenarios.

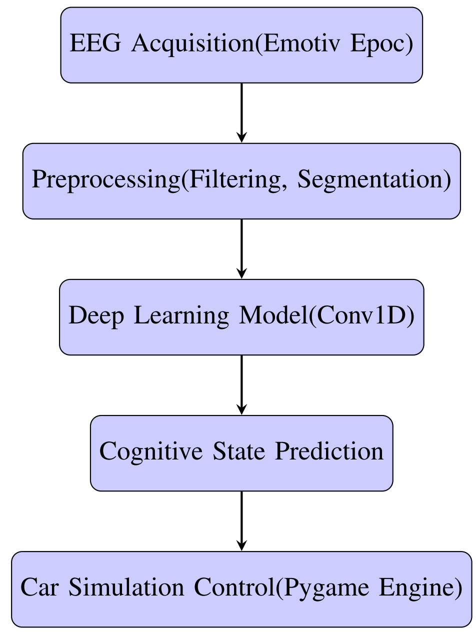

Fig. 1. System flowchart showing the pipeline from EEG acquisition to car control via cognitive state classification.

XIII. COMPARATIVE ANALYSIS OF THE APPROACHES

To validate our model, we compared it with a number of traditional and deep learning approaches:

TABLE III
COMPARATIVE ACCURACY FOR THE APPROACHES

Model	**Accuracy**	**Approach**
Bird et al. (2018)	87.16%	Random Forest + Feature Selection
EEGNet (Lawhern et al.)	84.57%	Depthwise CNN
CNN-LSTM (Nandihal et al.)	85.68%	Hybrid Deep Learning
Our Model	84.93%	Conv1D + Raw EEG Band Input

A. Discussion

While our model did not perform as well as traditional models that utilize engineered features, there were some advantages:

1) End-to-end learning with minimal to no explicit preprocessing,
2) Real-time inference,
3) A model architecture that can scale as future work is done.

The lower accuracy for Drowsy detection suggests a need for more discriminative features or more generative hybrid approaches. Nonetheless, the model has competent performance and is applicable in a real-time context.

XIV. Results and Discussion

The car simulation was implemented in real time using Python's Pygame to present images of the output of the model in relation to varying cognitive states. The speed of the vehicle varies based on EEG-based predictions of the driver's cognitive states: Focused, Unfocused, and Drowsy.

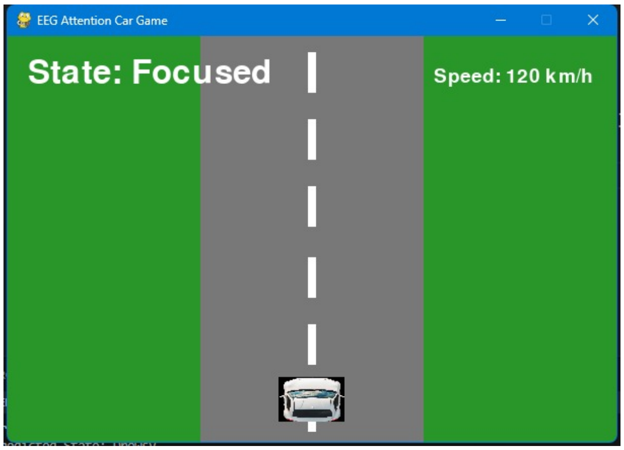

Fig. 2. Simulation output for **Focused** state (Speed: 120 km/h).

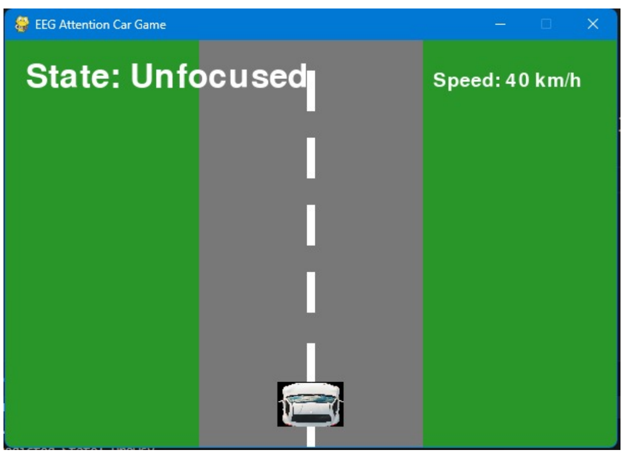

Fig. 3. Simulation output for **Unfocused** state (Speed: 40 km/h).

These images visibly illustrate how cognitive predictions are telegraphed to driving behavior in real-time. When the subject was attentive, the car maintained the appropriate speed. As attention waned or sleepiness increased, the system slowly decreased speed or paused the simulation altogether.

The simulation also gives visual and quantitative feedback on driver performance, offering researchers the opportunity to see the immediate impact that cognitive states have on driving behavior. The system clearly shows end-to-end functionality from brain signal acquisition to action output as a result of the integration of the EEG model and the simulation. In addition, the framework can be expanded to include multiple vehicle dynamics, experimental conditions, or additional cognitive features to further improve realism. The system facilitates real-time logging of speed, reaction times, and predicted cognitive states to be used as performance evaluations or for adaptive training. Beyond logging and data analysis, this setup provides a platform to perform experiments into intervention strategies (e.g., auditory or visual alerts) when the driver demonstrates signs of fatigue or inattention.

Fig. 4. Simulation output for **Drowsy** state (Vehicle stopped).

XV. Limitations and Future Work

Future research could consider using SMOTE or focal loss for class imbalance.

- **Class Imbalance:** Future research could consider using SMOTE or focal loss.
- **Dataset Generalization:** There is a need for evaluation across datasets.
- **Signal Quality:** More effective artifact rejection and adaptive filtering systems.
- **Personalization:** More can be done to add calibration or transfer learning per-user.

Future work might include multimodal systems (EEG + eye tracking + heart rate), transformer or GNN-based models, and embedding on devices to apply in a real-time driver assistance system.

XVI. Conclusion

This article describes a Conv1D-based EEG classification framework applied in a real-time car simulator. The framework is successfully able to classify Focused, Unfocused, and Drowsy mental states and adaptively control the vehicle in response. This framework serves as a scalable and practical means of developing neuroadaptive systems in both driving and interactive environments.

The results demonstrate the feasibility and benefits of coupling non-invasive brain monitoring and feedback in real-time, potentially increasing safety and performance. The framework aligns cognition state detection with real-time feedback in an interactive way, which could lead to the design of intelligent driver assistance systems, adaptive training and exercise platforms, and potentially individual neurofeedback applications.

The framework's modular design further encourages integration with other physiological signals or vehicle dynamics in order to create a richer and responsive experience.

Acknowledgment

The authors thank the contributors of the open EEG datasets and open-source tools such as TensorFlow, Pygame, and the Emotiv SDK.

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